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COMMON QUARTERLY EXAMINATION - 2026

T

Standard XII

Reg.No.

MATHEMATICS

Time : 3.00 hrs **Part – I** **Marks : 90**

I. Choose the correct answer. **20 x 1 = 20**

1. If A is a 3 x 3 non–singular matrix such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then BB^T =
   a) A b) B c) I_3 d) B^T
2. If A^TA^{-1} is symmetric, then A^2 is _____.
   a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$
3. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is _____.
   a) 1 b) 2 c) 3 d) 4
4. The product of all four values of $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is
   a) –2 b) –1 c) 1 d) 2
5. The principal argument of $(\sin 40^\circ + i\cos 40^\circ)^5$ is
   a) -110° b) -70° c) 70° d) 110°
6. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is
   a) –2 b) –1 c) 1 d) 2
7. If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive root, if and only if
   a) $a \geq 0$ b) $a > 0$ c) $a < 0$ d) $a \leq 0$
8. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if k satisfies
   a) $|k| \leq 6$ b) $k = 0$ c) $|k| > 6$ d) $|k| \geq 6$
9. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y$ is equal to _____.
   a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π
10. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is _____.
    a) [1,2] b) [–1,1] c) [0,1] d) [–1,0]
11. $\sin^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x)$ = _____.
    a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$
12. If $\sin^{-1}\frac{x}{5} + \operatorname{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is _____.
    a) 4 b) 5 c) 2 d) 3
13. The radius of the circle passing through the point (6,2) two of whose diameter are x + y = 6 and x + 2y = 4 is
    a) 10 b) $2\sqrt{5}$ c) 6 d) 4

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14. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is
a) 3 b) –1 c) 1 d) 9

15. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
a) 2ab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

16. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11,2), the coordinates of the other end are
a) (–5,2) b) (2,–5) c) (5, –2) d) (–2,5)

17. The angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k})$ and the plane $\vec{r}.(\hat{i} + \hat{j}) + 4 = 0$ is
a) 0° b) 30° c) 45° d) 60°

18. The distance between the planes $x + 2y + 3z = 0$ and $2x + 4y + 6z + 7 = 0$ is
a) $\frac{\sqrt{7}}{2\sqrt{2}}$ b) $\frac{7}{2}$ c) $\frac{\sqrt{7}}{2}$ d) $\frac{7}{2\sqrt{2}}$

19. Volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\hat{i} + 14\hat{j} + 10\hat{k}$, $14\hat{i} - 10\hat{j} - 6\hat{k}$ and $2\hat{i} + 4\hat{j} - 2\hat{k}$ is
a) 700 b) 500 c) 720 d) 520

20. Which one is meaningful ?
a) $(\vec{a} \times \vec{b}) \times (\vec{b}.\vec{c})$ b) $\vec{a} \times (5 + \vec{b})$ c) $(\vec{a}.\vec{b}) \times (\vec{c}.\vec{d})$ d) $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d})$

Part - II

II. Answer any Seven questions. (Q.No.30 is compulsory) 7 x 2 = 14

21. If A is symmetric, prove that adj A is also symmetric.

22. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

23. Show that $(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}$ is purely imaginary

24. Find the modulus and principal argument of $\sqrt{3} - i$.

25. If α, β and γ are the roots of the polynomial equation $ax^3 + bx^2 + cx + d = 0$, find the value of $\sum \frac{\alpha}{\beta\gamma}$ in terms of the coefficients.

26. Find the principal value of $\cot^{-1}(\sqrt{3})$

27. Find the equation of the ellipse with foci (± 2, 0), vertices (±3, 0)

28. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$, find $(\vec{a} \times \vec{b}) \times \vec{c}$.

29. Find the angle between the following lines: $2x = 3y = -z$ and $6x = -y = -4z$

30. Prove that $\begin{vmatrix} \hat{i} + \hat{j}, \hat{j} + \hat{k}, \hat{k} + \hat{i} \end{vmatrix} = 2$

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Part - III

III. Answer any Seven questions. (Q.No.40 is compulsory) 7 x 3 = 21

31. Solve the following system of linear equations by matrix inversion method:
$2x + 5y = -2,\ x + 2y = -3$ Ex 1.3 Q.No (1)(i)

32. Find the rank of $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$ Ex 1.2 Q.No (2)(i)

33. Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$ into rectangular form. Eg 2.4

34. Find the square root of $6 - 8i$ Eg 2.17

35. Find a polynomial equation of minimum degree with rational coefficients, having $2i + 3$ as a root. Ex 3.2 Q.No (3)

36. Find the domain of $\sin^{-1}(2 - 3x^2)$. Eg 4.4

37. If the equation $3x^2 + (3 - p)xy + qy^2 - 2px = 8pq$ represents a circle, find p and q. Also determine the centre and radius of the circle. Eg 5.1 (12)

38. Prove by vector method that an angle in a semi-circle is a right angle. Ex 6.1

39. Find the coordinates of the point where the straight line
$\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + t(3\hat{i} + 4\hat{j} + 2\hat{k})$ intersects the plane $x - y + z - 5 = 0$ Eg 6.56

40. Find the value of $\cos^{-1}\cos 5$. Ex 4.5 (1) (creative)

Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) Investigate the values of λ and μ the system of linear equations
$2x + 3y + 5z = 9,\ 7x + 3y - 5z = 8,\ 2x + 3y + \lambda z = \mu$, have
(i) no solution (ii) a unique solution (iii) an infinite number of solutions. Ex 1.6 Q.No (3)

(OR)

b) Solve the following system of linear equations by Cramer's rule:
$\frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0,\ \frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0,\ \frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$ Ex 1.4 (3)

42. a) If $2\cos\alpha = x + \frac{1}{x}$ and $2\cos\beta = y + \frac{1}{y}$ show that

i) $xy - \frac{1}{xy} = 2i\sin(\alpha + \beta)$ ii) $\frac{x^m}{y^n} - \frac{y^n}{x^m} = 2i\sin(m\alpha - n\beta)$

(OR)

b) If z_1, z_2 and z_3 are three complex numbers such that $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$

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43. a) Solve: $(x-5)(x-7)(x+6)(x+4) = 504$ Ex 3.4 (1)

(OR)

b) Solve the following equation: $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ Eg 3.28.

44. a) Solve: $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ Eg 4.28

(OR)

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x,y,z < 1$, then show that

$x^2 + y^2 + z^2 + 2xyz = 1$ Eg 4.22

45. a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4 m when it is 6m away from the point of projection. Finally it reaches the ground 12 m away from the starting point. Find the angle of projection. Ex 5.5 Q.No. 9

(OR)

b) Find the centre, foci, and eccentricity of the hyperbola

$11x^2 - 25y^2 - 44x + 50y - 256 = 0$ Eg 5.26

46. a) Find the equation of the circle passing through the points (1,1) ,(2,–1) and (3,2) . Eg 5.10

(OR)

b) Find the foot of the perpendicular drawn from the point (5,4,2) to the line Ex 6.5 Q.No (7)

$\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$. Also, find the equation of the perpendicular.

47. a) Using vector method, prove that $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

(OR)

b) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to plane $\vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = 8$ Ex 6.7 (5) Q.No